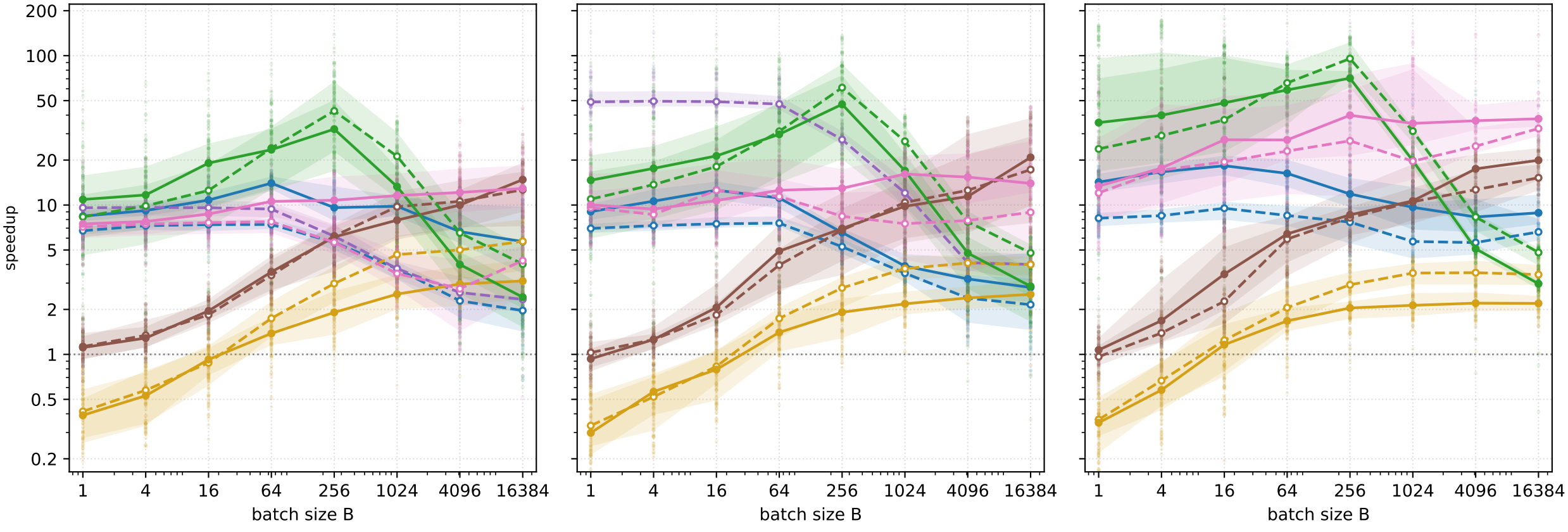

forward kinematics
frame Jacobians
frame Hessians
speedup
200
100
50
20
10
5
2
1
0.5
0.2
1
4
16
64
256
1024
4096
16384
batch size B
MuJoCo XLA fp32 (n=67)
MuJoCo XLA fp64 (n=67)
CPU idealized fp32 (n=67)
CPU idealized fp64 (n=67)
CPU + transfer fp32 (n=67)
CPU + transfer fp64 (n=67)
CusADi fp32 (n=67)
CusADi fp64 (n=67)
frax fp32 (n=67)
frax fp64 (n=67)
MuJoCo Warp fp32 (n=60–67)